INCIDENT FILE P2W-0412 · PRELIMINARY REPORT

A PDF Was Asked to Become a Word Document

Findings of the review into the formatting event of Tuesday afternoon

Summary

At 14:07 a routine request was made to turn a two-page PDF back into an editable Word document. Within seconds the headings had separated from their paragraphs, a table had become eleven text boxes, and the page numbers were found in the middle of a sentence. No fonts were harmed. Several margins were not recovered.

This report sets out what happened, why it keeps happening, and what the converter in use at the time was never going to understand: a PDF records where every fragment of text sits on the page and nothing about what the text means.

> **Key finding.** The structure of the document was never lost. It was never in the file. It has to be rebuilt from geometry: which pieces share a baseline, which gaps are columns, which lines are rules.

Timeline

Time	Event	Severity
14:07	PDF opened in the usual converter	None
14:07	Two-column layout read straight across, left line then right line	High
14:08	Bullet points converted to the letter o and a small square	Moderate
14:08	Table borders drawn as separate floating lines	High
14:11	User resumed typing the whole thing out by hand	Severe

Table 1. Sequence of events, local time.

Findings

1. Headings were identified by size, not by position, so a bold sentence was promoted to chapter level.
2. Columns were never detected, so every second line belonged to a different paragraph.
3. List markers were kept as text, so editing a list meant renumbering it by hand.
4. Running headers and page numbers were treated as body text on every page.

Contributing factors

- Text boxes are an easy way to make a PDF look right and a hard way to make it editable
- Justified text produces wide gaps that look like table columns
  - Gaps that repeat down the page are columns
  - Gaps that do not repeat are just spaces
- Scanned pages contain no text at all, only a picture of text

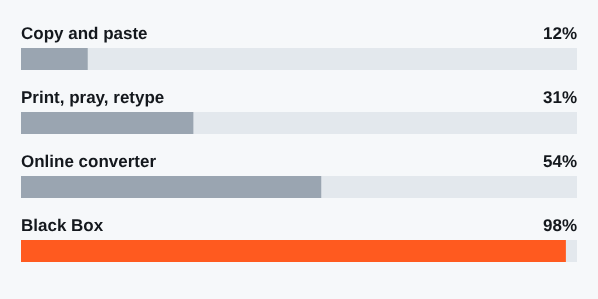


Figure 1. Share of original formatting kept after conversion, by method.

Recommendations

Read each page as geometry. Group fragments into lines by baseline, split lines where gaps are wide, and look for gaps that line up from one row to the next.

Rebuild reading order before writing anything. A column is finished when its last line ends, not when the page does, so the left column comes before the right.

Write real structure: heading styles, list numbering, table cells, page-number fields. A document that only looks right is the problem that started this report.

Keep a ledger. Every character that went in should come out, and the converter should say so when one does not.